

Xuan Qi

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Education

Tsinghua University

Beijing, China

Undergraduate, Institute for Interdisciplinary Information Sciences (IIIS, Yao Class;
directed by Andrew C. Yao, Turing Award laureate 2000), GPA: 3.85/4.00

Sept 2022 – Present

- **Coursework:** Natural Language Processing (A⁺), Economics and Computation (A⁺), Introduction to Programming in C/C++ (A⁺), Causal and Statistical Learning (A⁺), Mathematics for Computer Science (A), Introduction to Computer Systems (A), Abstract Algebra (A), Probability and Statistics (A), Introduction to Optimization Theory (A), Algorithm Design (A), Machine Learning (A), Game Theory (A)
- **Selected awards:** Yao Award; Tsinghua University Comprehensive Excellence Scholarship; First Prize in the Provincial Division of the Chinese Mathematics Competitions; First Prize in the Provincial Division of the National Olympiad in Informatics in Provinces; First Prize in the Provincial Division of the Chinese High School Mathematics League.

Work Experience

Research Intern, Cognitive Computation Group

Philadelphia, PA, USA

University of Pennsylvania (Advisor: Prof. Dan Roth)

May 2025 – Aug 2025

- Worked on the *DataProphet* project, analyzing cross-modal transferability of supervision signals across reasoning tasks and, based on these findings, performing real-data reweighting and synthetic-data selection. The resulting data subsets were substantially higher quality than other strong baseline methods, leading models trained under the same budget to clearly outperform baselines and revealing systematic patterns of supervision transfer that improve multimodal LLM data synthesis and training strategies.

Algorithm Research Intern

Beijing, China

Reinforce Omega (AI Agent Start-up)

Sept 2025 – Feb 2026

- Designed and iterated on *AI agent* workflows, including tool-use policies, tool-creation pipelines, and skill document generation/management to improve robustness and reusability.
- Built context and memory management mechanisms to support long-horizon planning, error recovery, and multi-step execution across heterogeneous tools.
- Contributed to the agent platform backend, including designing agent-friendly websites, improving web/GUI agent capabilities, and creating benchmarks for evaluating agent performance.

AI Agent Research Intern

Beijing, China

Xiaohongshu HI Lab (ACE Top Intern Program)

Mar 2026 – Present

- Investigated existing *AI-for-Research* benchmarks, reproduced prior state-of-the-art systems, and explored new agent designs for advancing the performance frontier of research-oriented agents.
- Built *agentic systems for research* with a focus on two core problems: generating valuable, feasible research ideas and conducting persuasive validation via theoretical reasoning and experimental analysis.
- Explored system-level improvements spanning agent workflow design, memory management, MCP and skill integration, and auxiliary model training to enhance long-horizon planning, reasoning, and execution.

Publications

* indicates equal contribution.

Demystifying Supervision Data Generalization in Multimodal LMs

2026

Xuan Qi, Luxi He, Dan Roth, Xingyu Fu.

ICLR 2026. [https://openreview.net/forum?id=iYMZKz5BGz&referrer=%5BAuthor%20Console%5D\(%2Fgroup%3Fid%3DICLR.cc%2F2026%2FConference%2FAuthors%23your-submissions\)](https://openreview.net/forum?id=iYMZKz5BGz&referrer=%5BAuthor%20Console%5D(%2Fgroup%3Fid%3DICLR.cc%2F2026%2FConference%2FAuthors%23your-submissions))

Alita: Generalist Agent Enabling Scalable Agentic Reasoning with Minimal Predefinition and Maximal Self-Evolution

2025

Jiahao Qiu*, Xuan Qi*, Tongcheng Zhang*, Xinzhe Juan, Jiacheng Guo, Yifu Lu, Yimin Wang, Zixin Yao, Qihan Ren, Xun Jiang, Xing Zhou, Dongrui Liu, Ling Yang, Yue Wu, Kaixuan Huang, Shilong Liu, Hongru Wang, Mengdi Wang

In submission to top conference. <https://arxiv.org/abs/2505.20286> 

Alita-G: Self-Evolving Generative Agent for Agent Generation	2025
Jiahao Qiu*, Xuan Qi *, Hongru Wang*, Xinzhe Juan, Yimin Wang, Zelin Zhao, Jiayi Geng, Jiacheng Guo, Peihang Li, Jingzhe Shi, Shilong Liu, Mengdi Wang In submission to top conference. https://arxiv.org/abs/2510.23601 	
A Survey of Self-Evolving Agents: On Path to Artificial Super Intelligence	2025
Huan-ang Gao*, Jiayi Geng*, Wenyue Hua*, Mengkang Hu*, Xinzhe Juan*, Hongzhang Liu*, Shilong Liu*, Jiahao Qiu*, Xuan Qi *, ... (Co-first authors list ordered alphabetically by last name.) TMLR. https://arxiv.org/abs/2507.21046 	
Shallow Preference Signals: Large Language Model Aligns Even Better with Truncated Data?	2025
Xuan Qi *, Jiahao Qiu*, Xinzhe Juan, Yue Wu, Mengdi Wang ACL 2025 Workshop. https://arxiv.org/abs/2505.17122 	
Difficulty-Based Preference Data Selection by DPO Implicit Reward Gap	2025
Xuan Qi , Rongwu Xu, Zhijing Jin In submission to top conference. https://arxiv.org/abs/2508.04149 	
DebateQA: Evaluating Question Answering on Debatable Knowledge	2024
Rongwu Xu*, Xuan Qi *, Zehan Qi, Wei Xu, Zhijiang Guo Conditionally accepted by TACL. https://arxiv.org/abs/2408.01419 	
Truthful Dataset Valuation by Pointwise Mutual Information	2024
Shuran Zheng*, Xuan Qi *, Rui Ray Chen*, Yongchan Kwon, James Zou ICLR 2025 Workshop. https://arxiv.org/abs/2405.18253 	
Learning Agent Routing from Early Experience	2026
Yimin Wang*, Jiahao Qiu*, Xuan Qi , Xinzhe Juan, Jingzhe Shi, Zelin Zhao, Hongru Wang, Shilong Liu, Mengdi Wang In submission to top conference.	
On Path to Multimodal Historical Reasoning: HistBench and HistAgent	2025
Jiahao Qiu*, Fulian Xiao*, Yimin Wang*, Yuchen Mao*, Yijia Chen*, Xinzhe Juan, Siran Wang, Xuan Qi , Tongcheng Zhang, Zixin Yao, et al. In submission to top conference. https://arxiv.org/abs/2505.20246 	
AgentDistill: Training-Free Agent Distillation with Generalizable MCP Boxes	2025
Jiahao Qiu*, Xinzhe Juan*, Yimin Wang*, Ling Yang*, Xuan Qi , Tongcheng Zhang, Jiacheng Guo, Yifu Lu, Zixin Yao, Hongru Wang, Shilong Liu, Xun Jiang, Liu Leqi, Mengdi Wang In submission to top conference. https://arxiv.org/abs/2506.14728 	
HDI4D: A Hand-Deformable Object Interaction Dataset for Soft Body Hanging	2024
Pengyang Shi*, Xiaoxi Liang*, Xiaomeng Xu, Xuan Qi , Tongzhou Du, Yi Li	

Research Experience

Wang Lab@Princeton AI for Accelerating Invention	<i>Remote</i>
Princeton University (Advisor: Prof. Mengdi Wang)	Nov 2024 – Feb 2026
- Co-led the development of <i>Alita</i>, a general agent with minimal pre-definition and maximal evolution that evolves via dynamic Model Context Protocol (MCP) creation and achieved top-1 performance on the GAIA benchmark. - Extended the <i>Alita</i> framework to <i>Alita-G</i>, which systematically abstracts and curates MCP tools to transform generalist agents into domain experts and provide a principled pathway from generalist capabilities to reusable, domain-specific competence. - Served as co-first author of <i>A Survey of Self-Evolving Agents</i>, mapping the emerging landscape of self-evolving architectures and agentic systems. - Served as co-first author of <i>Shallow Preference Signals: Large Language Model Aligns Even Better with Truncated Data?</i>, showing that truncated preference signals can effectively steer model behavior and revealing that models often rely on shallow heuristics for alignment. - Contributed to <i>AgentDistill</i>, exploring training-free distillation via Model Context Protocols to transfer reasoning and tool-use abilities between agents.	
EconCS Lab	<i>Beijing, China</i>
Tsinghua University, IIIS (Advisor: Assistant Prof. Shuran Zheng)	Jul 2024 – Jan 2025

- Developed a metric based on pointwise mutual information to value datasets in machine learning and encourage honest data contributions, and carried out theoretical analyses and experiments to validate its effectiveness in *Truthful Dataset Valuation by Pointwise Mutual Information*.

Jinesis AI Lab

Remote

University of Toronto (Advisor: Assistant Prof. Zhijing Jin)

Aug 2024 – Feb 2025

- Investigated data efficiency under Prof. Zhijing Jin, contributing to *Difficulty-Based Preference Data Selection by DPO Implicit Reward Gap* by using implicit reward gaps to guide preference data selection.

Xu Wei Research Group

Beijing, China

Tsinghua University, IIIS (work with Master's student Rongwu Xu)

Mar 2024 – Jul 2024

- Initiated the *DebateQA* project, constructing a dataset of over 2,000 debatable questions on controversial topics and designing an evaluation framework that combined perplexity measures with model-based judging to evaluate LLMs on debatable knowledge.

The 2nd Robot Dog Competition

Beijing, China

Tsinghua University, Department of Automation

Apr 2024 – Jul 2024

- Trained Xiaomi's CyberDog2 to play soccer: trained the robot dog to perform locomotion and dribbling in a simulated environment using reinforcement learning, combined the two using a rule-based approach, and utilized interference frequency optimization, default pose adjustment, and PD control parameter tuning to solve the sim-to-real gap; developed evaluation methods based on this dataset to assess the performance of debatable QA.
- Our project got the Best Technology Award of the 2nd Robot Dog Competition organized by the Department of Automation.

3DVICI Lab

Beijing, China

Tsinghua University, IIIS (Advisor: Assistant Prof. Yi Li)

Aug 2023 – Oct 2024

- Spearheaded the data pipeline for *HDI4D: A Hand-Deformable Object Interaction Dataset for Soft Body Hanging*, designing robust data-collection pipelines, managing complex annotations, and empirically validating models to support robotic understanding of deformable object interactions.

Awards and Honors

- Yao Award (10 out of 96), 2025.
- Tsinghua Friends – Huawei Scholars First-Class Scholarship (10 out of 96), 2025.
- Science and Technology Innovation Excellence Scholarship, Tsinghua University (10 out of 96), 2025.
- Outstanding Volunteer and Public Service Scholarship, Tsinghua University (5 out of 96), 2025.
- Tsinghua University Comprehensive Excellence Scholarship (15 out of 96, awarded to those with outstanding performance), 2023, 2024.
- Tsinghua Friends – Zhang Mingwei Scholarship (10 out of 96), 2023.
- Tsinghua Friendship – Huang Yicong and Spouse Scholarship (10 out of 96), 2024.
- Best Technology Award of the 2nd Robot Dog Competition organized by the Department of Automation (1 out of 50 teams), 2024.
- First Prize in the Provincial Division of the Chinese Mathematics Competitions, 2023.
- Outstanding Student of Shandong Province (High School), 2021.
- First Prize in the Provincial Division of the Chinese High School Mathematics League, 2021.
- First Prize in the Provincial Division of the National Olympiad in Informatics in Provinces (NOIP), 2020.
- Second Prize in the National High School Physics Competition, 2021.
- Second Prize in the Provincial Division of the National High School Chemistry League, 2021.
- Second Prize in the Provincial Division of the National Olympiad in Informatics in Provinces (NOIP), 2019.

Leadership and Service

- Reviewer for ACL Rolling Review 2025 (ARR 2025), ICLR 2026, and AAAI 2026.
- Served as the team leader of the Tsinghua University “Love for Alma Mater” event's Jining Confucius School team, and as a key volunteer for the Tsinghua University Jining Regional Admissions Group in 2023 and 2024.
- Participated in multiple rural teaching-support programs organized by the Tsinghua University SAEPA (Student Association for Education and Public Affairs), including classroom teaching and one-on-one mentoring for high school students in under-resourced areas.
- Tsinghua University Q&A Forum Senior Volunteer.

- Tsinghua University Three-Star Zijing Volunteer.
- Served as the Class Publicity Officer, responsible for designing and promoting class activities.

Skills

- Languages: Chinese, English.
- Programming Languages: Python, C/C++, JavaScript, etc.
- Tools and Platforms: Git, L^AT_EX, SQL, Xiaomi CyberDog2, Raspberry Pi, etc.
- Other: Passionate Go player; achieved third place in the *San Yi Cup* National Dan Promotion Tournament and ranked in the top ten multiple times in the Shandong Youth and Children Championships.